

Problems on Trains - Formulas

1. km/hr to m/s conversion:

$$a \text{ km/hr} = (a * 5/18) \text{ m/s}$$

2. m/s to km/hr conversion:

$$a \text{ m/s} = (a * 18/5) \text{ km/hr}$$

Formulas for finding Speed, Time and Distance

Time taken by a train of length l metres to pass a pole or standing man or a signal post is equal to the time taken by the train to cover l metres.

Time taken by a train of length l metres to pass a stationery object of length b metres is the time taken by the train to cover $(l + b)$ metres.

Suppose two trains or two objects bodies are moving in the same direction at u m/s and v m/s, where $u > v$, then their relative speed is $= (u - v)$ m/s.

Suppose two trains or two objects bodies are moving in opposite directions at u m/s and v m/s, then their relative speed is $= (u + v)$ m/s.

If two trains of length a metres and b metres are moving in opposite directions at u m/s and v m/s, then:

The time taken by the trains to cross each other =	$(a + b)$	sec.
	$(u + v)$	

If two trains of length a metres and b metres are moving in the same direction at u m/s and v m/s, then:

The time taken by the faster train to cross the slower train = $(a + b)/(u - v)$ sec.

If two trains (or bodies) start at the same time from points A and B towards each other and after crossing they take a and b sec in reaching B and A respectively, then:

$$(A's \text{ speed}) : (B's \text{ speed}) = (\sqrt{b} : \sqrt{a})$$

Problems on Trains - General Questions

1. A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train?

120 metres

180 metres

324 metres

150 metres

2. A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is:

45 km/hr

50 km/hr

54 km/hr

55 km/hr

3. The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is:

200 m

225 m

245 m

250 m

4. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

1 : 3

3 : 2

3 : 4

None of these

5. A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?

120 m

240 m

300 m

None of these

6. A train 240 m long passes a pole in 24 seconds. How long will it take to pass a platform 650 m long?

65 sec

89 sec

100 sec

150 sec

7. Two trains of equal length are running on parallel lines in the same direction at 46 km/hr and 36 km/hr. The faster train passes the slower train in 36 seconds. The length of each train is:

50 m

72 m

80 m

82 m

8. A train 360 m long is running at a speed of 45 km/hr. In what time will it pass a bridge 140 m long?

40 sec

42 sec

45 sec

48 sec

9. Two trains are moving in opposite directions @ 60 km/hr and 90 km/hr. Their lengths are 1.10 km and 0.9 km respectively. The time taken by the slower train to cross the faster train in seconds is:

36

45

48

49

10. A jogger running at 9 kmph alongside a railway track in 240 metres ahead of the engine of a 120 metres long train running at 45 kmph in the same direction. In how much time will the train pass the jogger?

3.6 sec

18 sec

36 sec

72 sec

11. A 270 metres long train running at the speed of 120 kmph crosses another train running in opposite direction at the speed of 80 kmph in 9 seconds. What is the length of the other train?

230 m

240 m

260 m

320 m

None of these

12. A goods train runs at the speed of 72 kmph and crosses a 250 m long platform in 26 seconds. What is the length of the goods train?

230 m

240 m

260 m

270 m

13. Two trains, each 100 m long, moving in opposite directions, cross each other in 8 seconds. If one is moving twice as fast the other, then the speed of the faster train is:

30 km/hr

45 km/hr

60 km/hr

75 km/hr

14. Two trains 140 m and 160 m long run at the speed of 60 km/hr and 40 km/hr respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is:

9

9.6

10

10.8

15. A train 110 metres long is running with a speed of 60 kmph. In what time will it pass a man who is running at 6 kmph in the direction opposite to that in which the train is going?

5 sec

6 sec

7 sec

10 sec

16. How many seconds will a 500 metre long train take to cross a man walking with a speed of 3 km/hr in the direction of the moving train if the speed of the train is 63 km/hr?

25

30

40

45

17. Two goods train each 500 m long, are running in opposite directions on parallel tracks. Their speeds are 45 km/hr and 30 km/hr respectively. Find the time taken by the slower train to pass the driver of the faster one.

12 sec

24 sec

48 sec

60 sec

18. Two trains are running in opposite directions with the same speed. If the length of each train is 120 metres and they cross each other in 12 seconds, then the speed of each train (in km/hr) is:

10

18

36

72

19. Two trains of equal lengths take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 metres, in what time (in seconds) will they cross each other travelling in opposite direction?

10

12

15

20

20. A train 108 m long moving at a speed of 50 km/hr crosses a train 112 m long coming from opposite direction in 6 seconds. The speed of the second train is:

48 km/hr

54 km/hr

66 km/hr

82 km/hr

21. Two trains are running at 40 km/hr and 20 km/hr respectively in the same direction. Fast train completely passes a man sitting in the slower train in 5 seconds. What is the length of the fast train?

23 m

23	$\frac{2}{9}$	m
----	---------------	---

27	$\frac{7}{9}$	m
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29 m

22. A train overtakes two persons who are walking in the same direction in which the train is going, at the rate of 2 kmph and 4 kmph and passes them completely in 9 and 10 seconds respectively. The length of the train is:

45 m

50 m

54 m

72 m

23. A train overtakes two persons walking along a railway track. The first one walks at 4.5 km/hr. The other one walks at 5.4 km/hr. The train needs 8.4 and 8.5 seconds respectively to overtake them. What is the speed of the train if both the persons are walking in the same direction as the train?

66 km/hr

72 km/hr

78 km/hr

81 km/hr

24. A train travelling at 48 kmph completely crosses another train having half its length and travelling in opposite direction at 42 kmph, in 12 seconds. It also passes a railway platform in 45 seconds. The length of the platform is

400 m

450 m

560 m

600 m

25. Two stations A and B are 110 km apart on a straight line. One train starts from A at 7 a.m. and travels towards B at 20 kmph. Another train starts from B at 8 a.m. and travels towards A at a speed of 25 kmph. At what time will they meet?

9 a.m.

10 a.m.

10.30 a.m.

11 a.m.

Problems on Trains - Data Sufficiency 1

Directions to Solve

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question. Read the both statements and

- Give answer (A) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question.
- Give answer (B) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question.
- Give answer (C) if the data either in Statement I or in Statement II alone are sufficient to answer the question.
- Give answer (D) if the data even in both Statements I and II together are not sufficient to answer the question.
- Give answer (E) if the data in both Statements I and II together are necessary to answer the question.

1.

What is the speed of the train whose length is 210 metres?	
I.	The train crosses another train (Howrah Express/12869) of 300 metres length running in opposite direction in 10 seconds.
II	The train crosses another train (Howrah Express/12869) running in the same direction at the speed of 60 km/hr in 30 seconds.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

What is the length of a running train crossing another 180 metre long train running in the opposite direction?
--

I. The relative speed of the two trains was 150 kmph.

II. The trains took 9 seconds to cross each other.
--

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

What is the length of a running train?
--

I. The train crosses a man in 9 seconds.
--

II. The train crosses a 240 metre long platform in 24 seconds.
--

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

Problems on Trains - Data Sufficiency 2

Directions to Solve

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

1.

What is the speed of the train?	
I.	The train crosses a signal pole in 18 seconds.
II.	The train crosses a platform of equal length in 36 seconds.
III.	Length of the train is 330 metres.

I and II only

II and III only

I and III only

III and either I or II only

Any two of the three

2.

What is the speed of the train?	
I.	The train crosses a tree in 13 seconds.
II.	The train crosses a platform of length 250 metres in 27
III.	The train crosses another train running in the same direction in

I and II only

II and III only

I and III only

Any two of the three

None of these

Time and Distance - Formulas

1. Speed, Time and Distance:

$$\text{Speed} = \left(\frac{\text{Distance}}{\text{Time}} \right), \text{ Time} = \left(\frac{\text{Distance}}{\text{Speed}} \right), \text{ Distance} = (\text{Speed} \times \text{Time}).$$

2. km/hr to m/sec conversion:

$$x \text{ km/hr} = \left(x \times \frac{5}{18} \right) \text{ m/sec.}$$

3. m/sec to km/hr conversion:

$$x \text{ m/sec} = \left(x \times \frac{18}{5} \right) \text{ km/hr.}$$

4. If the ratio of the speeds of A and B is $a : b$, then the ratio of

the times taken by them to cover the same distance is $\frac{1}{a} : \frac{1}{b}$ or $b : a$.

5. Suppose a man covers a certain distance at x km/hr and an equal distance at y km/hr. Then,

the average speed during the whole journey is $\left(\frac{2xy}{x + y} \right)$ km/hr.

Time and Distance - General Questions

1. A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour?

3.6

7.2

8.4

10

2. An aeroplane covers a certain distance at a speed of 240 kmph in 5 hours. To cover the same distance in $1\frac{2}{3}$ hours, it must travel at a speed of:

300 kmph

360 kmph

600 kmph

720 kmph

3. If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is:

50 km

56 km

70 km

80 km

4. A train can travel 50% faster than a car. Both start from point A at the same time and reach point B 75 kms away from A at the same time. On the way, however, the train lost about 12.5 minutes while stopping at the stations. The speed of the car is:

100 kmph

110 kmph

120 kmph

130 kmph

5. Excluding stoppages, the speed of a bus is 54 kmph and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?

9

10

12

20

6. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is:

1 hour

2 hours

3 hours

4 hours

7. A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.

220 km

224 km

230 km

234 km

8. The ratio between the speeds of two trains is 7 : 8. If the second train runs 400 km in 4 hours, then the speed of the first train is:

70 km/hr

75 km/hr

84 km/hr

87.5 km/hr

9. A man on tour travels first 160 km at 64 km/hr and the next 160 km at 80 km/hr. The average speed for the first 320 km of the tour is:

35.55 km/hr

36 km/hr

71.11 km/hr

71 km/hr

10. A car travelling with $\frac{5}{7}$ of its actual speed covers 42 km in 1 hr 40 min 48 sec. Find the actual speed of the car.

$17 \times \frac{6}{7}$

25 km/hr

30 km/hr

35 km/hr

11. In covering a distance of 30 km, Abhay takes 2 hours more than Sameer. If Abhay doubles his speed, then he would take 1 hour less than Sameer. Abhay's speed is:

5 kmph

6 kmph

6.25 kmph

7.5 kmph

12. Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?

8 kmph

11 kmph

12 kmph

14 kmph

13. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the cars is:

2 : 3

3 : 2

3 : 4

4 : 3

14. A farmer travelled a distance of 61 km in 9 hours. He travelled partly on foot @ 4 km/hr and partly on bicycle @ 9 km/hr. The distance travelled on foot is:

14 km

15 km

16 km

17 km

15. A man covered a certain distance at some speed. Had he moved 3 kmph faster, he would have taken 40 minutes less. If he had moved 2 kmph slower, he would have taken 40 minutes more. The distance (in km) is:

35

$36 \frac{2}{3}$

$37 \frac{1}{2}$

40

Time and Distance - Data Sufficiency 1

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1.

Two towns are connected by railway. Can you find the distance	
I.	The speed of the mail train is 12 km/hr more than that of an
II.	A mail train takes 40 minutes less than an express train to cover

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

2.

The towns A, B and C are on a straight line. Town C is between A and B. The distance from A to B is 100 km. How far is A from C?

I. The distance from A to B is 25% more than the distance from C

II. The distance from A to C is $\frac{1}{4}$ of the distance C to B.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

3.

Two cars pass each other in opposite direction. How long would

The sum of their speeds is 135 km/hr.

The difference of their speed is 25 km/hr.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

Both I and II are not sufficient to answer

Both I and II are necessary to answer

4.

How much time did X take to reach the destination?
--

I. The ratio between the speed of X and Y is 3 : 4.

II. Y takes 36 minutes to reach the same destination.

I alone sufficient while II alone not sufficient to answer

II alone sufficient while I alone not sufficient to answer

Either I or II alone sufficient to answer

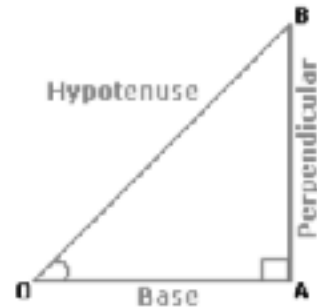
Both I and II are not sufficient to answer

Both I and II are necessary to answer

Height and Distance - Formulas

1. Trigonometry:

In a right angled Δ OAB, where $\angle BOA = \theta$,



i. $\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}} = \frac{AB}{OB}$

ii. $\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}} = \frac{OA}{OB}$

iii. $\tan \theta = \frac{\text{Perpendicular}}{\text{Base}} = \frac{AB}{OA}$

iv. $\operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{OB}{AB}$

v. $\sec \theta = \frac{1}{\cos \theta} = \frac{OB}{OA}$

vi. $\cot \theta = \frac{1}{\tan \theta} = \frac{OA}{AB}$

2. Trigonometrical Identities:

1. $\sin^2 \theta + \cos^2 \theta = 1$.

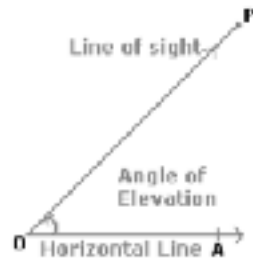
2. $1 + \tan^2 \theta = \sec^2 \theta$.

3. $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.

3. Values of T-ratios:

θ	0°	$(\pi/6)$	$(\pi/4)$	$(\pi/3)$	$(\pi/2)$
		30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined

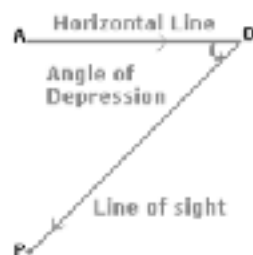
4. Angle of Elevation:



Suppose a man from a point O looks up at an object P, placed above the level of his eye. Then, the angle which the line of sight makes with the horizontal through O, is called the **angle of elevation** of P as seen from O.

$\therefore$ Angle of elevation of P from O = $\angle AOP$.

5. Angle of Depression:



Suppose a man from a point O looks down at an object P, placed below the level of his eye, then the angle which the line of sight makes with the horizontal through O, is called the **angle of depression** of P as seen from O.

Height and Distance - General Questions

1. Two ships are sailing in the sea on the two sides of a lighthouse. The angle of elevation of the top of the lighthouse is observed from the ships are 30° and 45° respectively. If the lighthouse is 100 m high, the distance between the two ships is:

173 m

200 m

273 m

300 m

2. A man standing at a point P is watching the top of a tower, which makes an angle of elevation of 30° with the man's eye. The man walks some distance towards the tower to watch its top and the angle of the elevation becomes 60° . What is the distance between the base of the tower and the point P?

43 units

8 units

12 units

Data inadequate

None of these

3. The angle of elevation of a ladder leaning against a wall is 60° and the foot of the ladder is 4.6 m away from the wall. The length of the ladder is:

2.3 m

4.6 m

7.8 m

9.2 m

4. An observer 1.6 m tall is 203 away from a tower. The angle of elevation from his eye to the top of the tower is 30° . The height of the tower is:

21.6 m

23.2 m

24.72 m

None of these

5. From a point P on a level ground, the angle of elevation of the top tower is 30° . If the tower is 100 m high, the distance of point P from the foot of the tower is:

149 m

156 m

173 m

200 m

6. The angle of elevation of the sun, when the length of the shadow of a tree 3 times the height of the tree, is:

30°

45°

60°

90°